MARK SCHEME · TEACHER COPY

10BR · Chapter 3: Hardware

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 70

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 70 marks

Question 1 · May/June 2024 Paper 12, original Q5 (a),(b) [8 marks]

Question	Answer	Marks
1(a)	**One** mark for each correct term in the correct place: • address • memory address register // MAR • random access memory // RAM • memory data register // MDR • data • current instruction register // CIR • control unit // CU	7
1(b)	A **list** of (machine code) **commands** that can be processed by the CPU	1

Question 2 · May/June 2024 Paper 12, original Q7 (a),(b) [4 marks]

Question	Answer	Marks
2(a)(i)	**A**	1
2(b)(i)	An IP address that has numerical values separated by dots that follows the format with a max value of 255 in any xxx xxx.xxx.xxx.xxx Example: 10.245.3.99	1
2(b)(ii)	Any **two** from: • 128-bit // 16 bytes • Hexadecimal • Separated by colons • Characters in groups of 4 • Has 8 groups of characters • Double colons can be used for sets of (consecutive) zeros (only once)	2

Question 3 · May/June 2024 Paper 12, original Q9 (a),(b) [8 marks]

Question	Answer	Marks

Total: 70 marks

3(a)	Any **two** from: • To store data/files **permanently** • … otherwise, data/programs would need to be downloaded/entered/installed every time the computer was used • To allow **software** to be **installed** on the computer • For the creation of virtual memory	**2**
3(b)	Any **six** from (**Max 3** for one type only): • Solid-state has no moving parts • … optical does have moving parts • Solid-state is faster to **access** data • … optical is slower to **access** data • Solid-state uses less power to operate • … optical uses more power to operate • Solid-state runs quietly • … optical makes more noise • Solid state is more durable/robust • … optical is less durable/robust • Solid state storage has a large storage capacity • … optical has a much smaller limit to its storage capacity • Solid-state is more expensive **per GB** of data • … optical is cheaper **per GB** of data • Solid-state stores data onto silicon chips/transistors using logic (floating/control/NAND/NOR) gates to control the movement of electrons • … optical stores data by using lasers to burning pits and lands onto a disk	**6**

Question 4 · May/June 2025 Paper 12, original Q1 (a),(b) [2 marks]

Question	Answer	Marks
4(a)	Any **one** from: • ROM • Cache	**1**

Total: 70 marks

4(b)	Any **one** from: • It is volatile storage // Data is lost when the power is turned off. • It needs to be **regularly** replaced by other data // The data **regularly** changes // Data needs to be **constantly** updated	**1**

Question 5 · May/June 2025 Paper 12, original Q2 (a) [1 marks]

Question	Answer	Marks
5(a) i)	A	**1**

Question 6 · May/June 2025 Paper 12, original Q3 (c) [11 marks]

Question	Answer	Marks
6(a)(i)	Any **one** from: • Inkjet • Laser	**1**

6(a)(ii)	**One** mark from: • Serial • Parallel **One** mark from: • Half-duplex • Full-duplex • Simplex Any **four** from (for descriptions matching transmission types given): **serial** • Serial would send bits in order // serial uses only one wire. • … so won't be skewed // less likely to have errors • Serial transmission speed would be adequate. **parallel** • Parallel would transmit data faster. • … as multiple bits are sent at the **same time** // … as multiple wires are used. • For parallel, chance of **skewing/errors** would be low as **short distance** transmission only required. **half/full duplex** • To allow data to be sent in both directions • … so any interrupts/notifications for errors can be sent back to the computer. **simplex** • Data **only** needs to be sent one direction // Data transmission doesn't need to be two-way. • … as the printer may not need to send errors back to the computer	**6**
6(a)(iii)	• A parity bit is added to each **byte.** • … to make the number of 1s/0s even // that will be 1 if the number of 1s/0s is odd // that will be 0 if the number of 1s/0s is even. • The number of 1s/0s in each byte is counted **after transmission.** • If any bytes have an odd number of 1s/0s an error is detected	**4**

Total: 70 marks

Question 7 · May/June 2025 Paper 12, original Q4 (a),(b) [8 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>7(a)</td><td>One mark for each correct component or description

<table>
<tr><th>Component</th><th>Description</th></tr>
<tr><td>Control unit // CU</td><td>It sends signals to all the components in the CPU to manage the flow of data through the CPU.</td></tr>
<tr><td>Arithmetic and logic unit // ALU</td><td>It carries out all the arithmetic and logic operations in the CPU.</td></tr>
<tr><td>Cache</td><td>It stores frequently used data/instructions</td></tr>
<tr><td>Program counter (PC)</td><td>It stores the address of the next instruction to be fetched</td></tr>
<tr><td>Clock</td><td>It controls the number of fetch-decode-execute (FDE) cycles that are performed per second.</td></tr>
<tr><td>Memory data register // MDR</td><td>It stores data immediately before it is transmitted to RAM and immediately after it is received from RAM.</td></tr>
</table></td><td>6</td></tr>
<tr><td>7(b)</td><td>Two from:

• An embedded system is designed to perform a dedicated/limited/single function // computer can be used to perform many different functions.
• An embedded system has dedicated hardware // computer has hardware that can be used by other devices.
• An embedded system has software that is not easily updated/reprogrammed // software can easily be updated/reprogrammed on the computer.
• An embedded system has a microprocessor // A computer has a CPU.
• An embedded system can be part of/built into a larger device // A computer is normally standalone</td><td>2</td></tr>
</table>

Question 8 · May/June 2026 Paper 11, original Q2 (a),(b),(c) [11 marks]

Question	Answer	Marks	Guidance
8	Examiners must award **whole marks** (not half marks, or other fractions).		

8(a)(i)	**One** from: • Program counter // PC • Memory Address register // MAR	**1**	Mark the first answer given.
8(a)(ii)	**Three** from: • The instruction is sent/copied to the current instruction register/CIR/control unit/CU // The CIR/CU receives the instruction // The CIR stores/holds the instruction … • … from the memory data register/MDR • The **control unit/CU decodes** the instruction • An instruction set is used (to decode the instruction) • The instruction is decoded/converted into the Opcode and Operand	**3**	BOD data in place of instruction MP1 – BOD instruction register/IR in place of CIR MP1 – this is given for the instruction being sent to the CIR/control unit. Allow if it is coming from an incorrect or inaccurate place e.g. MAR/PC MP3 – It must clearly state the CU is decoding the instruction
8(a)(iii)	**One** from: • Arithmetic logic unit // ALU • Accumulator // ACC • Control unit // CU • (Data/address/control) bus • Memory data register // MDR • Memory address register // MAR • Program counter // PC • Random access memory // RAM	**1**	Mark the first answer given. Register alone is NE Do not award CIR Award appropriate registers that are outside the specification e.g. Index register (IX) and status register (SR)
8(b)	**Two** from: • It can execute **more** instructions/FDE/FE/clock cycles **simultaneously** // **four** instructions/FDE/FE/clock cycles can be executed **simultaneously** rather than **two** • It can execute **more** instructions/FDE/FE/clock cycles **per second** • … as it has **double the number** of cores // It has **twice as many** cores // It has **more** cores // It has **4** cores instead of **2** cores	**2**	Tasks/processes/code/information/commands/actions/data is TV for instructions MP1 and 2 – BOD clock ticks/pulses in place of cycles

8(c)(i)	**Two** from: For example: • Touchscreen • Microphone • Keyboard/keypad • Button • Sensor • (Digital) camera	**2**	Mark the first answer given. Award any appropriate example of a sensor. Sensor can only be awarded once. Do not allow onscreen keyboard as this is software. Do not allow touchpad as it is not relevant to context.
8(c)(ii)	**Two** from: For example: • Screen // touchscreen • Speaker • LED // light • Actuator	**2**	Mark the first answer given. Display alone is NE for screen.

Question 9 · May/June 2026 Paper 11, original Q4 (a) [1 marks]

Question	Answer	Marks	Guidance
9 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
9(a)	**One** from: • To **input data** // **interact** with/**navigate** the computer/interface • To control/move the **cursor** on the screen • To act as a pointing device/pointer • To allow the user to **click/select/choose/drag** (items on the screen) // by example • To **scroll** through different pages/a page	**1**	MP1 – 'it is an input device' is NAQ MP4 – BOD 'open' in place of click etc

Question 10 · May/June 2026 Paper 12, original Q1 (a),(d) [2 marks]

Question	Answer	Marks	Guidance

Total: 70 marks

10	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.		
10(a)	A	**1**	If more than one box ticked, no mark can be awarded.
10(b)	**One** from: • Accelerometer • Light • Temperature • Proximity • Infra-red • Acoustic • Magnetic field • Moisture • Humidity • Pressure	**1**	Mark first answer given. Motion, sound, touch and heart rate are TV. If an answer is not in the list but seems viable, please refer to your TL. Do not accept by description, for example it measures light.

Question 11 · May/June 2026 Paper 12, original Q2 (a),(b),(c),(d) [8 marks]

Question	Answer	Marks	Guidance
11	Examiners must award **whole marks** (not half marks, or other fractions).		
11(a)	**One** from: • To process/execute data/instructions • To perform/carry out/run/process the fetch – decode – execute/FDE/FE cycle	**1**	Tasks/programs/code/information in place of instructions is TV. MP2 – BOD do/does the FDE cycle
11(b)	**Two** from: • Program counter // PC • Current instruction register // CIR // instruction register // IR • Accumulator // ACC	**2**	Mark first answer given. Do not accept MAR or MDR as they are given in the question. Accept correct registers that are not on the spec e.g. index register // IX or status register // SR

11(c)	**Three** from: • The MAR **stores/holds/contains** the address/location/pointer // address is **sent/copied** to MAR … • … for the value/data that is to be fetched/in RAM • The MDR **stores/holds/contains** the value/data // data is **sent/copied** to the MDR	**3**	MP2 and 3 - BOD instruction in place of value/data
11(d)	**Two** from: • 3 billion • … is the (maximum) number of instructions / FDE cycles / FE cycles / clock cycles processed **per second**	**2**	Tasks/programs/code/information is TV for instructions MP1 – accept 3 billion by calculation e.g. 3×10^9 or 3 000 000 000 MP2 – BOD clock ticks/pulses in place of cycles.

Question 12 · May/June 2026 Paper 12, original Q3 (f) [3 marks]

Question	Answer	Marks	Guidance
12	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
12(a)(i)	**Two** from: • It uses electromagnets • It has platters (to store data) • (Platters are) divided into tracks and sectors • It has a read/write head/arm	**2**	Mark whole answer. MP1 – uses magnets is TV, must be electromagnet
12(a)(ii)	Solid-state // solid-state drive // SSD	**1**	Mark first answer given.

Question 13 · May/June 2026 Paper 12, original Q6 (c) [3 marks]

Question	Answer	Marks	Guidance
13	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Total: 70 marks

Annotation	Meaning
	Correct point
	Incorrect point
	Unclear response
FT	Follow through
REP	Repetition
I	Ignore
BOD	Benefit of doubt given
NBOD	Benefit of doubt not given
TV	Content of response too vague
NAQ	Not answered question
Annotation	**Meaning**
	Omission
	Section not relevant
	Section incorrect
Highlighter	Information copied from the text
SEEN	Page or response seen by examiner

13(a)(i)	**One** from • The app data won't take up **storage** space on their device // There will be more **storage** space on the device for other files • They don't need to download the data each time they get a new device • The data can be accessed from any device they use // The data is still accessible if their device breaks/is lost	**1**	Mark whole answer. Do not accept answers relating to not needing to maintain hardware or security, or it having unlimited storage. These are not relevant to the scenario given. MP3 – Can be accessed anywhere is TV. They could access it anywhere if it was stored on their phone.
13(a)(ii)	**Two** from: • Need an internet connection to access the data stored in the cloud // If internet is slow/unstable app could be difficult to use • If the cloud storage is down/broken/fails the app cannot be used/access to data is lost • Relies on a third party to keep any (personal) data secure/safe/private that is provide to the app // (personal) data stored by the app could be less secure/safe/private • Access speed to the data may be slower/more limited (than stored locally)	**2**	Mark whole answer. Do not accept answers relating to cost of cloud storage. This is not relevant to the scenario. MP1 – Wi-Fi/network connection is TV MP4 – BOD transmission speed may be slower